

Ethical Issues in Video Games: A Large-Scale Empirical Analysis of Gamers' Reviews

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The video game industry has experienced unprecedented growth, transforming games into complex socio-technical systems with cultural, educational, and economic significance. As their influence expands, ethical issues—ranging from addiction to censorship, fairness, and privacy—have become critical for both players and developers. Within the field of Software Engineering (SE), and more specifically Video Game SE (VGSE), these concerns are particularly salient, as they intersect with design, governance, and community interaction. When these aspects are poorly managed, communities tend to be particularly resentful, leading to repercussions for game developers. To understand this issue, we investigate how ethical issues manifest in video games through a large-scale analysis of user reviews from Steam. Building on existing ethical frameworks and research on game ethics, we propose an *ethical framework for VGSE* and evaluate it through a semi-automated qualitative analysis of over 320 000 reviews using Leximancer. Our findings reveal a wide range of concerns spanning all stages of the game lifecycle, with recurring themes of fairness, accessibility, addiction, censorship, and privacy. By organizing these concerns temporally and linking them to responsible actors, we provide a lifecycle- and actor-oriented framework that bridges general-purpose ethical models with domain-specific VGSE practices, offering actionable guidance for responsible game development.

Content Warning: This paper includes examples of content coming from real video games reviews that could be considered offensive.

CCS Concepts: • **Software and its engineering** → **Extra-functional properties**;

Additional Key Words and Phrases: Software Engineering, Video Game Development, Ethics and Fairness

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1 Introduction

Research consistently shows that video games—especially those with strong narratives or moral choices—can shape players' attitudes, empathy, and moral reasoning [21, 25, 33, 40, 43]. For example, narrative games and those with moral dilemmas promote self-examination, empathy, and contemplation of moral values, which can translate into real-world awareness and attitude change [25, 43].

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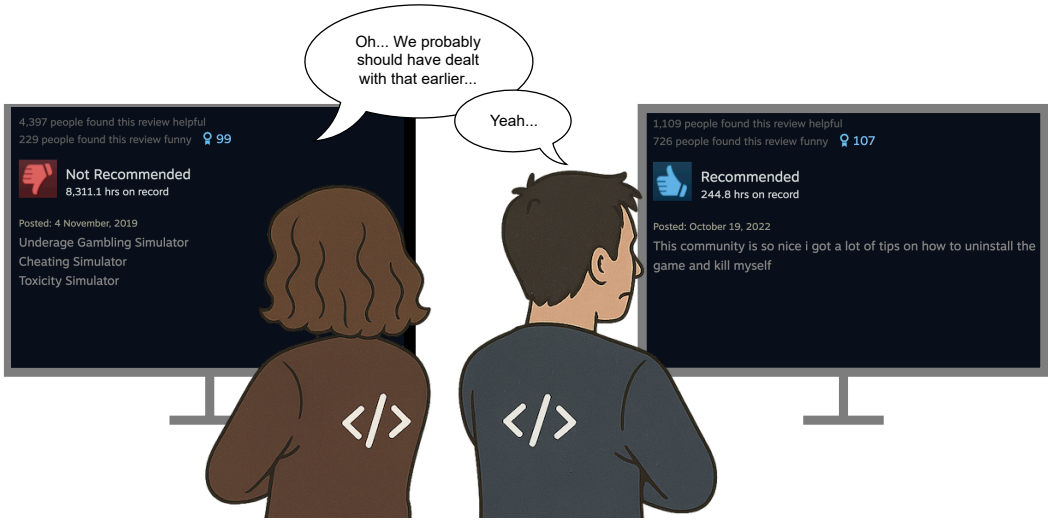


Fig. 1. Examples of two reviews from the most played game on Steam.

In addition, games that support social interaction, even when they are focused on violent behaviors [21]—can improve the attitude of players towards prosocial and cooperative behaviors [33, 40].

Among the values that video games can convey, there is a growing awareness—also within the software engineering (SE) community [36]—of ethical and moral aspects [25, 43]. Prior research has shown that video games can influence these dimensions in both positive and negative ways, and that multiple ethical facets are affected by specific elements of game design. For example, excessive violence can either foster aggressive attitudes [21] or increase players' awareness of immoral behaviors [25], while the proliferation of monetization systems raises concerns related to fairness, transparency, and gambling-like practices [24, 32]. These are only two among several ethical issues, and it is important to emphasize that their impact extends beyond social consequences for players: they also affect the companies themselves, which may—often unintentionally—suffer severe reputational damage as a result of misguided design choices. For example, big companies like Epic Games and Microsoft have faced major lawsuits for intentionally designing addictive systems targeting minors, using operant conditioning and variable reward schedules to boost engagement and monetization.¹ If we then consider that we are referring (only) to ethical issues in product design rather than in corporate structure, it becomes clear that the topic in question is of primary importance in the general context of video game software engineering (VGSE).

☹ Motivating Example.

Figure 1 presents two real user reviews from Steam for the platform's most-played game. The negative review (on the left) reports concerns related to gambling, cheating, and community toxicity—issues that more than 4000 other players endorsed as helpful, indicating broad agreement with the problems raised. Although many reviews appear positive, as illustrated by the example on the right, they are often written sarcastically and may actually convey

¹Video Game Addiction Lawsuit Settlements: <https://www.lawsuit-information-center.com/video-game-addiction-lawsuits.html>

strong criticism, such as frustration with toxic community behavior. This “positive” review also resonated with more than 1000 users, further demonstrating collective awareness of these problems. Together, these examples serve as warning signals for developers: player reviews frequently highlight urgent and recurrent issues that demand attention, as even seemingly positive feedback can mask severe concerns. This reinforces the importance of systematically and qualitatively analyzing game reviews to identify ethical and technical problems early and guide responsive maintenance before player trust declines.

For the reasons outlined above, identifying and addressing such issues within a game is of primary importance. Nonetheless, doing so is far from straightforward: on one hand, these problems are conceptually complex and nontrivial, requiring the creation of classification frameworks, such as PEGI² and ESRB,³ which, however, often prove insufficient [10, 17]. On the other hand, once ethical issues emerge, it is rarely immediate to “refactor” or address them, since video game development is inherently chaotic and involves a wide range of professionals, introducing organizational barriers that make any corrective action time-consuming [15, 22]. For this reason, in recent years, the SE research community has increasingly leveraged diverse data sources—such as Reddit discussions, online reviews, and postmortems—to investigate these and similar phenomena [36, 38]. In particular, while reviews have been extensively analyzed in the context of mobile and general-purpose applications [44, 49], they have rarely been considered as a means to detect ethical concerns in video games. We argue that this is a significant gap, as player reviews provide a direct and authentic perspective through which ethical issues can be identified and, if properly interpreted and translated into actionable insights, can help development teams improve their products. Moreover, such information is also crucial to safeguard players themselves, promoting safer, fairer, and more responsible gaming environments.

Building on the above-mentioned premises, and with the aim of opening a promising new research line within the VGSE domain, this work investigates the presence and contextualization of ethical issues in video game reviews. To address this, we formulated the following research question (RQ).

❓ RQ — *What types of ethical issues concerning players’ experiences with video games emerge from the content of player reviews?*

To answer the RQ and reach the objective, we conducted a two-phase study: (1) the definition of a preliminary framework of ethical issues in video games, grounded in existing literature; and (2) the evaluation of its applicability through a semi-automated qualitative study. For the second phase, we constructed and analyzed a large-scale dataset comprising more than 320 000 Steam reviews, processed through a supervised pipeline employing Leximancer [42],⁴ a qualitative analysis tool for text mining and thematic analysis that is well established in other research communities [16, 20, 48] but remains virtually unexplored within SE.

Our findings depict an intriguing picture. The proposed framework not only finds manifestations across numerous reviews of different games, but these ethical issues also further decompose into a hierarchical structure that often reveals connections between seemingly unrelated concerns. Moreover, the reviews frequently contain detailed information that could, in principle, be incorporated by automated tools into maintenance and evolution processes—thus supporting, within the VGSE context, a research direction that has long received attention in SE. Building on our results, it becomes plausible to envision a future in which automated systems extract data and

²Pan European Game Information: <https://pegi.info/it>

³Entertainment Software Rating Board: <https://www.esrb.org>

⁴Leximancer website: <https://www.leximancer.com>

construct pipelines that help companies identify and resolve ethical problems, safeguard players by preventing the reinforcement of morally questionable behaviors, and gradually build organizational know-how that informs new developers throughout the game development process—ultimately increasing their success rate.

The paper is structured as follows. Section 2 introduces the related work that forms the foundation for understanding and contextualizing the results of this contribution. Section 3 outlines the ethical framework for VGSE and the way we developed it. Section 4 describes the research design, focusing on the methodology, analysis tools, data collection, and analysis procedures. Section 5 presents the qualitative findings and the discussion of the results for each theme in the ethical framework, while Section 6 interprets the findings taking a broad perspective and looking for connections between them, providing actionable recommendations for practitioners and outlining directions for future research. Finally, Section 7 concludes the paper.

2 Related Work

This section describes the related work that is the foundation for the contribution.

2.1 Ethical Issues in Video Games

Ethical concerns in video games have received sustained attention in scholarly debates, as games have evolved into complex socio-technical artefacts that both reflect and shape cultural values. Existing studies emphasize how video games mirror broader moral and social dynamics, often reproducing real-world inequalities or raising questions of fairness and responsibility.

A large body of research has examined issues of representation, diversity, and stereotyping in games. Shaw [39] argues that video games frequently replicate social inequalities, offering limited and often stereotypical portrayals of women. Consalvo [6] notes that even when diversity is present, minority characters are typically constrained to secondary or stereotypical roles. Empirical analyses confirm that female characters are sexualized and underrepresented, reinforcing harmful gender norms [2, 4]. Leonard [35] similarly demonstrates how race is commodified in video games, as characters of color are often depicted through reductive or exoticized stereotypes.

The impact of violent content has long been one of the most contested ethical questions in the gaming literature. Dowsett and Jackson [12] find a correlation between violent gameplay and aggressive tendencies, while Adachi and Willoughby [1] argue that competitiveness and game difficulty may explain much of this effect. Meta-analyses by Ferguson [19] indicate that the connection between violent games and real-world aggression is weak, highlighting the influence of mediating factors such as player personality. Nonetheless, scholars emphasize the ethical responsibility of developers in providing transparent content warnings and effective parental controls [30].

The introduction of microtransactions and loot boxes has opened an extensive debate on the ethical boundaries of monetization. Griffiths [23] and King and Delfabbro [31] argue that these mechanisms often rely on psychological triggers resembling gambling, exploiting reward anticipation to encourage repeated spending. Loot boxes, in particular, have been shown to blur the distinction between play and wagering, especially among younger audiences [24]. These trends contribute to an ethical dilemma in which financial incentives may override player well-being. The convergence between gaming and gambling practices further reinforces this concern [9].

Finally, ethical complexity extends to the moral architecture of games themselves. Many titles allow players to make moral decisions, yet these are often simplified into binary oppositions such as “good” versus “evil”. This reductionist framing, as shown by Sicart [41] and Ferchaud and Oliver [18], limits the potential of games as spaces for moral reflection and meaningful ethical engagement. Nevertheless, Grizzard et al. [25] demonstrated in a study how players who find themselves

performing “evil” or immoral actions in games are then able to develop greater awareness of these dynamics, thereby improving the way they approach them.

Overall, the literature reveals a diverse and interconnected set of ethical challenges—ranging from representation and violence to monetization and player agency. While rating systems such as PEGI and ESRB provide basic consumer guidance, they remain insufficient to address the systemic ethical tensions embedded in the design and economy of contemporary video games [11, 17].

2.2 Ethical Frameworks in IT Systems

When studying contributions in terms of ethical frameworks in relation to video game from an SE perspective, the only work that we found in the SE community was that of Li and Aslam [36]. Moving to the general context of IT, two other contributions emerged, that is, the one from Tjikhoeri et al. [44] that is based on the work of Wright [49].

A foundational framework is the *Ethical Impact Assessment* (EIA) proposed by Wright [49]. Originally developed in the context of information technology policy and system design, the framework provides a structured methodology for identifying and assessing ethical issues associated with emerging technologies. Drawing upon the bioethical principles of Beauchamp and Childress—*respect for autonomy, non-maleficence, beneficence, and justice*—together with a dedicated focus on *privacy and data protection*, Wright’s model translates abstract ethical theory into actionable evaluative dimensions. It proposes a question-driven assessment process aimed at technology developers and policymakers to guide the examination of moral implications before deployment. Crucially, the framework emphasizes stakeholder engagement, transparency, and context-sensitive deliberation, positioning ethics as an integral component of socio-technical decision-making. This approach has since been widely recognized as a cornerstone for ethical evaluation in IT systems, serving as the conceptual foundation upon which later domain-specific adaptations have been built.

The second framework considered in our study is that of Tjikhoeri et al. [44], which investigates ethical concerns emerging from large-scale analyses of mobile app store reviews. The authors collected over five million reviews from diverse applications and manually annotated more than three thousand instances to identify recurring ethical themes. Their framework defines a taxonomy of sixteen ethical concerns—ranging from *privacy* and *transparency* to *censorship*, *identity theft*, and *content theft*—reflecting users’ lived experiences and moral perceptions of software behavior. These constructs are supported by both qualitative and quantitative evidence, bridging ethical theory with empirical data from end users. Conceptually, this framework draws inspiration from Wright’s Ethical Impact Assessment [49], adopting its core ethical values while adapting them to the domain of user-generated feedback. Whereas Wright’s model was intended as a normative assessment tool for policymakers and system designers, Tjikhoeri et al. operationalized those ethical principles into observable user concerns expressed in natural language. In doing so, they refined the original framework by simplifying overly abstract categories and introducing new ones—such as *cyberbullying*, *addictive design*, and *misinformation*—that capture emergent issues in contemporary digital ecosystems. This evolution transforms the EIA framework into a data-driven, user-centered taxonomy applicable to software evaluation and ethical monitoring at scale.

The third framework, and the one most directly relevant to our study, is the work of Li and Aslam [36], which introduces an exploratory analysis of ethical issues in video games based on large-scale Reddit discussions. Drawing from 19,843 posts across diverse gaming-related communities, their study combines qualitative coding and machine learning classification to identify, categorize, and automate the detection of ethical concerns expressed by players, developers, and related stakeholders. The resulting taxonomy encompasses seven primary categories—*violence*, *monetization*, *prejudice*, *addictive design*, *gambling*, *exclusion*, and *other moral dilemmas*—representing how ethical tensions materialize within game content, design practices, and community discourse. Conceptually,

the framework extends the empirical lineage of Tjikhoei et al. [44], who in turn had operationalized the one made by Wright [49]. While Tjikhoei et al. [44]. focused on the ethical dimensions of mobile applications through large-scale app reviews, Li and Aslam [36] contextualize those ethical constructs within the socio-technical ecosystem of video games. Their approach shifts the analytical focus from user–application relationships to community-mediated moral discourse, highlighting how players and developers negotiate issues of fairness, inclusion, and responsibility in interactive entertainment. Furthermore, by integrating qualitative thematic coding with supervised machine learning, the authors advance methodological rigor in the computational study of ethics in software engineering. This contribution provides a crucial empirical foundation for adapting ethical frameworks to the domain of VGSE.

❗ Novelty and Contribution.

We advance prior work on ethical frameworks in software and gaming by bridging the gap between general-purpose models for IT systems and mobile apps [44, 49] and domain-specific analyses of video games [36]. Whereas existing frameworks identify what ethical concerns exist, we contextualize them within the video game lifecycle—clarifying when they arise and who bears responsibility for addressing them. Methodologically, we complement the Reddit-based analysis from [36], which captures community-level moral discourse, with evidence from Steam reviews, where players directly communicate with developers about concrete design and governance issues. Conceptually, our framework moves beyond static taxonomies to provide actionable guidance for ethical game development and maintenance, translating players’ moral perceptions into lifecycle-aware responsibilities for practitioners.

3 Ethical Framework for VGSE

To investigate the presence and contextualization of ethical issues in video game reviews, it was first necessary to construct a conceptual framework capable of identifying and organizing the main ethical concerns relevant to the gaming domain. This section presents the framework development process, its theoretical foundations, and the constructs that compose it.

3.1 Ethical Framework Development Process

The first step of our research work involved building a framework that could be used to categorize ethical issues in video games reviews. The overall process aligned with the methodological principles of conceptual framework development [29] and integrative literature review [47]. Specifically, three theoretical sources—Tjikhoei et al. [44], Wright [49], and Li and Aslam [36]—were analyzed and merged through iterative collaborative sessions following a theory adaptation approach [28]. The resulting framework was subsequently refined and validated through expert workshops inspired by Delphi-like consensus techniques [27].

The first study, proposed by Tjikhoei et al. [44], provides a well-established and widely recognized classification of ethical issues in the IT context, which has been adopted in numerous subsequent works. The constructs identified in this contribution form the theoretical foundation for much of the literature on the topic and serve as an essential starting point for extending the analysis to the video game domain. These constructs were later examined in the context of mobile application reviews by the work of Wright [49]. The third work, a contribution from Li and Aslam [36], in contrast, introduces a qualitative analysis of ethical issues directly within the video game context. Although this contribution has not yet been quantitatively validated, it offers a specific and context-sensitive view of the phenomenon that proved valuable for adapting the framework to the VGSE domain. The decision to include it was also motivated by the intent to build upon

prior SE research while providing the first large-scale empirical evaluation of the framework for its practical applicability.⁵

Building upon the three theoretical frameworks previously discussed, we identified a set of constructs relevant to our study. The process initially involved two of the main authors, with subsequent review and validation provided by the remaining co-authors. In the first phase, the authors read and analyzed the reference literature and extracted the fundamental constructs. Then, through a series of collaborative meetings between the two authors, each construct was individually examined to assess its relevance within the video game context and to explore potential practical manifestations of the concept in this domain. This phase followed the methodological principles of conceptual framework development [29] and integrative literature synthesis [47], combining analytical rigor with interpretive reflection to ensure both theoretical grounding and contextual appropriateness. As a reference example, we report the construct labeled “Content Theft.” In the original formulation by Tjikhoei et al. [44] and Wright [49], this construct was defined as “Content from a user is stolen or used without permission from the original creator.” Following the aforementioned process, we reinterpreted and contextualized this definition within the domain of video games. Through iterative discussions and comparisons with empirical evidence from player reviews, we refined the construct to capture ethical concerns about situations in which user-generated content—such as mods, skins, level designs, or narrative ideas—is replicated, reused, or monetized without appropriate permission or attribution. Furthermore, our contextual adaptation extends the scope of the construct to include cases where developers themselves may integrate community creations into official releases without proper acknowledgment or compensation. In this way, the essence of the original construct is preserved, while its meaning is expanded and recontextualized to reflect ethical dynamics specific to video game ecosystems.

Once the individual analyses were completed, all constructs were compared to identify potential overlaps or redundancies, which were merged when necessary. All decisions were reached unanimously. The resulting set of constructs was then shared with the remaining co-authors, who provided additional feedback on possible empirical manifestations. Finally, the framework was presented and discussed in two internal workshops organized within the research group of the first three authors, involving approximately ten participants per session, including researchers active in the fields of Software Engineering and fairness in software development systems. This collaborative refinement phase followed the principles of interpretive synthesis and consensus validation [8, 27], ensuring that the final framework reflected both theoretical coherence and practical relevance. The outcome of this process is detailed in the following section.

3.2 Constructs of the Ethical Framework for VGSE

The final ethical framework for VGSE consists of constructs derived from prior literature on ethical issues in technology and video game contexts. Each construct represents a distinct ethical dimension that can manifest in player experiences and community interactions. While the framework was developed deductively, its application to large-scale review data led to minor empirical refinements, such as clarifying construct boundaries or adding relevant terminology, based on insights from the thematic analysis. In the following, we describe each construct, its conceptual boundaries, and typical ways it emerges within player reviews; the entire framework is summarized in Table 1.

Accessibility. This construct concerns situations where the application does not include people with special needs or disabilities. This concern can relate to the usability of the application itself or to a service it offers. This construct was elicited from the work of Wright [49] and the work

⁵A detailed description of the three studies and frameworks is provided in Section 2.2.

Table 1. Ethical issues constructs in the VGSE Framework.

Construct	Description
Accessibility	Refers to the absence of features that make games inclusive for players with disabilities, such as subtitles, color-blind modes, or adaptive controls.
Accountability	Highlights the lack of clear responsibility or support channels when users encounter problems like bugs, bans, or cheating.
Addiction / Gambling	Covers manipulative mechanics like loot boxes or time-gating that encourage compulsive usage or excessive spending.
Censorship	Involves unfair content removal or silencing of players by moderators or developers without clear justification.
Content Theft	Refers to the unauthorized use of user-created or third-party content, often by developers, without proper credit or compensation.
Discrimination / Monetization / Prejudice	Captures bias in systems or communities and unfair advantages tied to real-world spending (e.g., pay-to-win).
Cyberbullying	Concerns toxic behavior, harassment, or hate speech in in-game chats or community spaces.
Sustainability	Relates to resource-intensive games or poor optimization that causes unnecessary energy use or hardware strain.
Harmful Advertising	Refers to misleading or inappropriate ads, especially when targeted at vulnerable groups or embedded in inappropriate contexts.
Identity Theft	Covers impersonation or account hacking for in-game gain, rather than broader real-world identity fraud.
Inappropriate Content	Encompasses disturbing or violent content that may be upsetting or unsuitable for certain players.
Privacy	Relates to misuse or insecurity of user data, including overcollection, tracking, or account breaches.
Safety	Addresses health risks, doxxing, or real-world threats arising from use of the game or its community.
Scam	Covers financial or item-based deception, such as fake trades, fraudulent sellers, or tricking users into loss.
Misinformation	Concerns false claims about game features, updates, or mechanics, and unmoderated spread of inaccurate content.
Transparency	Involves unclear rules, monetization, or system mechanics, leaving players uncertain or mistrustful.
Social Solidarity / Exclusion	Captures exclusionary dynamics based on region, cost, or lack of support for marginalized groups.
Isolation and Substitution	Relates to feelings of social withdrawal due to games replacing real-life interaction.
Dignity	Involves content or behavior that humiliates, stigmatizes, or exploits players or groups.

of Tjikhoeri et al. [44]. *Example:* “The game lacks accessibility features like subtitles, colorblind modes, or remappable controls, making it difficult or impossible for disabled players to play.”

Accountability. This construct concerns situations where users encounter issues with the application or its services but cannot find the responsible company or developers to resolve the problem. This construct was elicited from the work of Wright [49] and the work of Tjikhoei et al. [44]. *Example:* “Players cannot get support for bugs, cheaters, or unfair bans, with no clear way to contact developers or get resolutions.”

Addiction / Addictive Design and Gambling. This construct concerns cases where users mention being addicted to the application or describe excessive use. It also includes gambling-like or manipulative design (e.g., loot boxes, excessive time-gates). This construct was elicited from the work of Li and Aslam [36] and the two works from Wright [49] and Tjikhoei et al. [44]. *Example:* “Games use loot boxes, time-gated events, and manipulative design that encourages compulsive play or overspending.”

Censorship. This construct concerns cases where the application deliberately hides certain information, or where certain users’ content or profiles are deliberately removed or demoted. This construct was elicited from the work of Wright [49] and the work of Tjikhoei et al. [44]. *Example:* “Game developers or moderators unfairly remove posts, ban players, or suppress specific discussions without clear justification.”

Content Theft. This construct concerns situations where content created by users—such as mods, skins, designs, or narrative ideas—is stolen, replicated, or used without proper permission or attribution. Such misuse can occur both among players and from developers themselves, for instance, when official game content incorporates community creations without acknowledgment or compensation. This construct was elicited from the work of Wright [49] and the work of Tjikhoei et al. [44]. *Empirical analysis showed that players mainly accused developers, not just other users, of appropriating community content or, more frequently, assets from other games, reinforcing perceptions of unethical corporate behavior. Example:* “Developers integrated a mod originally made by fans into the official release without crediting or rewarding the creators.”

Discrimination / Monetization (Pay-to-Win) / Prejudice. This construct concerns cases where users are discriminated against by the application, its services, or its community. It also includes situations where users have fewer functionalities available due to geographic restrictions or feel disadvantaged by pay-to-win mechanics. This construct was elicited from the work of Li and Aslam [36] and the two works from Wright [49] and Tjikhoei et al. [44]. *Example:* “Players face bias (sexism, racism) in chat or from systems, or feel the game is unfairly tilted toward players who spend money (‘pay to win’).”

Cyberbullying. This construct concerns situations where the platform’s community is harmful, abusive, or unhealthy, engaging in hateful communication via the application. This construct was elicited from the work of Wright [49] and the work of Tjikhoei et al. [44]. *Example:* “In-game chats or forums are toxic, with harassment, hate speech, or targeted abuse against players.”

Sustainability. This construct concerns users’ observations about the negative environmental impact of the application. This construct was elicited from the work of Wright [49] and the work of Tjikhoei et al. [44]. *Empirical findings revealed that players primarily expressed frustration with games that overheat systems, crash due to poor performance, or rely on unnecessarily resource-heavy infrastructure—indirectly pointing to sustainability issues without naming them as such. Example:* “Players complain about servers or hardware consuming excessive energy or games encouraging wasteful practices.”

Harmful Advertising. This construct concerns users noticing the presence of deceptive, misleading, or harmful advertisements within the application. This construct was elicited from the work of Wright [49] and the work of Tjikhoeri et al. [44]. *Empirical evidence revealed limited discussion of in-game advertisements, likely due to their lower prevalence in PC games compared to mobile platforms. Mentions were often conflated with issues of deceptive promotion about the game itself, which more closely aligned with the Scam construct. Example:* “Games show misleading or adult ads, including gambling or drug-related content, even in inappropriate contexts (like games for kids).”

Identity Theft. This construct concerns situations where someone is using another person’s identity within the application. This also applies to catfishing or creating fake profiles to deceive other users. This construct was elicited from the work of Wright [49] and the work of Tjikhoeri et al. [44]. *Empirical evidence suggests that identity-related concerns in videogames are relatively uncommon and tend to focus more on account hacking or impersonation for in-game advantage, rather than stealing real-world identities or creating fake profiles for broader deception. This likely reflects the more abstract, digital nature of user identities in gaming contexts. Example:* “Scammers impersonate other players or streamers, or use fake profiles to steal items or money.”

Inappropriate Content / Violence. This construct concerns the presence of content (other than advertisements) that may be disturbing to certain groups of people, including excessive violence, sexual content, or distressing imagery. This construct was elicited from the work of Li and Aslam [36] and the two works from Wright [49] and Tjikhoeri et al. [44]. *Empirical evidence showed that depictions of violence and disturbing content are often framed positively by players, particularly when aligned with genre expectations (e.g., horror or extreme combat games), complicating the boundary between entertainment and ethical concern. Example:* “The game contains graphic violence, sexualized characters, or other disturbing elements without adequate warnings or controls.”

Privacy. This construct concerns cases where users’ identity and data are not kept secure or are used for purposes other than those for which consent was given. It also includes cases where accounts are hacked. This construct was elicited from the work of Wright [49] and the work of Tjikhoeri et al. [44]. *Example:* “Games collect or share excessive player data, track user activity, or have security flaws leading to account breaches.”

Safety. This construct concerns cases where using the application has led to health issues or other safety risks. The concern can relate to the use of the application itself or its services. This construct was elicited from the work of Wright [49] and the work of Tjikhoeri et al. [44]. *Example:* “Players face stalking, doxxing, or physical harm linked to the game or its community.”

Scam. This construct concerns situations where users have been scammed or have encountered scammers through the application. Such scams usually involve deception for financial or material gain. This construct was elicited from the work of Wright [49] and the work of Tjikhoeri et al. [44]. *Example:* “Players lose money or in-game items due to fake trades, fraudulent sellers, or deceptive offers.”

Misinformation. This construct concerns the spread of false or misleading information through the application. This construct was elicited from the work of Wright [49] and the work of Tjikhoeri et al. [44]. *Empirical analysis showed that most references to misinformation concerned inaccurate or deceptive representations of game content, features, or updates. Mentions of user-generated misinformation, particularly fake reviews, were also frequent. Example:* “Players spread false claims about the game, its mechanics, updates, or related real-world topics, and developers fail to intervene.”

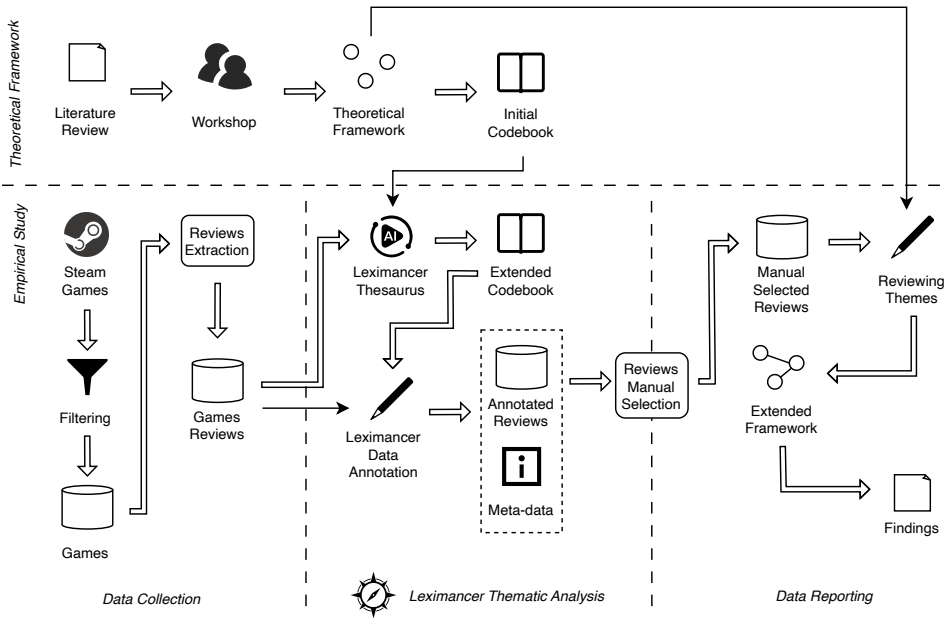


Fig. 2. Empirical research methods.

Transparency. This construct concerns situations where the motives, risks, or implications of using the application or its services are unclear to users. This construct was elicited from the work of Wright [49] and the work of Tjikhoeri et al. [44]. *Example:* “Game systems like loot boxes, monetization, or bans are not explained clearly, leaving players confused or mistrustful.”

Social Solidarity/Exclusion. This construct concerns cases where certain groups or individuals are unfairly excluded from participating in the application or its community. This can include cost barriers, regional restrictions, or lack of support for marginalized groups. This construct was elicited from the work of Li and Aslam [36] and the two works from Wright [49] and Tjikhoeri et al. [44]. *Example:* “Players feel excluded due to region-locks, overpriced content, or lack of community support for marginalized or new players.”

Isolation and Substitution. This construct concerns cases where users feel that the application contributes to their social isolation or substitutes for meaningful real-world interactions. This construct was elicited from the work of Wright [49] and the work of Tjikhoeri et al. [44]. *Example:* “Players feel that the game replaces face-to-face interaction, leading to withdrawal from social life or worsening loneliness.”

Dignity. This construct concerns situations where users feel that the application, its content, or its community undermines their dignity by humiliating, exploiting, or stigmatizing them or others. This construct was elicited from the work of Wright [49] and the work of Tjikhoeri et al. [44]. *Example:* “The game or its community includes content or behavior that humiliates or exploits players or NPCs (e.g., mockery of disabilities).”

4 Research Method

The goal of the empirical evaluation is to test the theoretical framework presented in the previous section by examining whether—and how—players discuss the identified ethical issues within video game reviews.

To achieve this goal, we designed a qualitative study based on *thematic analysis* [3, 8], in which—through a deductive approach—we investigate the presence and manifestation of the concepts outlined in the theoretical framework within player reviews. Supporting this analysis, the present work stands among the first in SE to employ Leximancer as a tool for qualitative inquiry, providing both an overview of the process and an initial account of the experience of using the tool for the analysis of SE-related artifacts.

Leximancer is a qualitative analysis software based on text mining and machine learning algorithms, designed to identify recurring concepts and their semantic relationships within large textual corpora [42]. Unlike traditional manual approaches, Leximancer automates part of the coding process and thematic map construction, enabling the detection of latent patterns in the data and the exploration of concept co-occurrences in a replicable and scalable manner.

In summary, the main phases of the study—reported in Figure 2—were as follows:

- (1) collection and pre-processing of the review dataset;
- (2) execution of the analysis using Leximancer; and
- (3) interpretation of the results and comparison with the reference literature.

The rest of this section provides detailed information on each of the steps mentioned above.

4.1 Data Collection

The dataset employed in this study consists of 326 440 individual entries, each representing a player review associated with one of the 24 701 video games considered. Each entry includes the textual content of the review along with relevant metadata, such as review polarity (positive or negative), game category, price, and Metacritic score. The dataset covers reviews posted between August 2021 and September 2025, with an average review length of approximately 60 words (340 characters). Overall, 82.31% of the reviews are positive and 17.69% negative. The collection includes both free and paid games (3381 and 8478 respectively), with an average price of 10.19 USD and an average Metacritic score of 80.13. The most represented genres include *Indie*, *Action*, *Adventure*, and *Simulation*.⁶

The dataset was constructed using a two-step approach: (1) gathering a comprehensive list of video games published on Steam, and (2) retrieving user reviews for those games via the official Steam Web API. For filtering, we chose to collect reviews of games published after 2020 for several reasons. First, in terms of impact, a 5-year window is reasonable to identify modern industry trends, as including older games might introduce outdated information into the sample. Second, this time frame makes the analysis feasible by limiting the amount of data to a manageable size, which would otherwise be overwhelming if extended beyond 5 years.

In the first phase, we queried the complete list of Steam applications via the dedicated public endpoint, retrieving metadata for all the available apps on the platform. Due to the high volume of apps on Steam, however, this endpoint only shares the *public name* and *appid*, i.e., a unique identifier for each app available on Steam. Hence, to collect the actual metadata for each game published in the last five years, we again queried the Steam Web APIs to get details for each singular app with the aim of filtering apps with two criteria. First, since the list also includes non-game content such as DLCs, software tools, and test builds, we filtered the results to retain only actual

⁶Ulterior information on the data can be found in our online appendix.

games. We applied multiple keyword-based heuristics to check that the app, even if not explicitly marked as “game” in the app type, was indeed a video game. Second, we checked that the release date was not earlier than five years ago. In cases where the API did not provide the release date, we performed an additional check by scraping the public web page.⁷ Since we were unable to retrieve all release dates because they were not available in the sources, we opted to keep the games and eventually discard them in the following step. Finally, we discarded entries with placeholder names, invalid or future release dates.

In the second phase, we used the Steam Reviews API endpoint to iteratively download reviews for each of the selected games. For every game, we retrieved up to 10 000 English-language reviews using a paginated cursor-based mechanism. To ensure the temporal consistency of the dataset, we discarded all games whose oldest available review was published more than five years before the data collection period. Each review record includes: a unique review ID, timestamp, user recommendation (positive/negative), free/purchased status, upvotes or downvotes, and the full review text.

After collecting the initial sample, a preliminary statistical analysis was conducted to examine data distribution and assess the need for further cleaning or reduction. Descriptive metrics such as polarity ratio, average review length, and genre frequency were computed to confirm dataset balance and representativeness across different categories of games.

Finally, the dataset was transformed and reformatted for import into *Leximancer*, which was used to perform the qualitative and conceptual analysis described in the following sections. Each review was treated as an individual text unit, and all metadata were preserved to enable potential triangulation between linguistic patterns and game attributes. The dataset is available in our online appendix [45].

4.2 Codebook Development and Thematic Analysis

To guide the thematic analysis of ethical concerns in videogame reviews, we adopted a deductive approach grounded in a theoretical framework of predefined ethical issues. From this framework, we derived an initial codebook, which included high-level themes (e.g., addiction, cyberbullying, privacy) and a preliminary list of keywords representing each theme based on their definition.⁸

To expand and empirically ground this codebook in our dataset, we employed *Leximancer*, a text analytics tool that extracts concepts by identifying high-frequency terms and their co-occurrence relationships [42]. Our use of *Leximancer* followed a two-phase process.

- (1) In the first phase, we allowed *Leximancer* to perform its default thesaurus learning, which automatically generates a set of semantically related terms for each concept based on statistical co-occurrence patterns across the dataset. This process provided a data-driven expansion of our initial keywords and revealed variations in vocabulary that might not have been captured manually. After this first run, we manually reviewed the resulting thesaurus for each concept, inspecting the top-ranked keywords suggested by *Leximancer*. We retained only those terms that we judged to be semantically coherent and meaningful in relation to our theoretical definitions. Additionally, we manually added synonyms, paraphrases, and related expressions that we found underrepresented. This led to a refined and curated codebook tailored to our analytical goals.

⁷**Ethical Statement.** All data was collected from publicly accessible web pages in compliance with Steam’s Terms of Use. No personal data was collected or inferred, and the scraping was performed responsibly, at low frequency and with user-agent identification, solely for academic research purposes.

⁸The initial codebook was created using the same methodology as the theoretical framework, as detailed in Section 3.

- (2) In the second phase, we execute Leximancer data annotation feature with thesaurus learning disabled, using only the manually refined codebook. This ensured that the analysis proceeded in a fully deductive manner, with concepts defined a priori and not further expanded by unsupervised learning.

In summary, at the conclusion of the second phase, Leximancer equipped us with a labeled dataset that highlighted themes aligned with our theoretical framework. The process predominantly relies on keyword matching, which naturally results in a number of false positives. This is understandable, as Leximancer is not intended to fully automate the process; instead, it provides a systematic way to identify potential themes that are valuable for qualitative discussions. Consequently, we conducted a final phase of data identification across all reviews to pinpoint examples that could bolster our discussion [42]. Although only a select number of reviews underwent detailed manual analysis, the entire dataset was utilized to enhance concept modeling and keyword validation, taking advantage of the linguistic and thematic variation within the corpus. Therefore, even though qualitative coding was applied selectively, the thematic analysis was based on patterns that emerged from the complete data set, thus reinforcing the validity and extent of the identified ethical themes.

We used two key functions of the Leximancer interface: (1) the concept map and (2) the indexing feature. The concept map provides a visual representation of the relationships among identified themes based on co-occurrence statistics, helping us to observe the distribution and clustering of ethical concerns across the dataset, while offering insights into broader topical structures and emergent macro-categories. The indexing feature allowed us to index each sentence in the dataset with the associated concepts based on the input codebook. This enabled us to perform targeted searches to retrieve reviews most aligned with specific ethical themes. By using Leximancer's search and filtering tools, we were able to explore reviews with and without keyword filters, finding representative and diverse examples of each ethical concern in user narratives. This iterative search process was conducted manually, continuing until we reached theoretical saturation, where additional reviews no longer provided new meaningful insights for a given theme.

4.3 Data Reporting

Once the final set of reviews for qualitative analysis was collected, the first author analyzed the data to identify any emerging concepts from the themes, such as their aggregation or additional horizontal dimensions between the themes. These relationships and new concepts were then discussed with the other authors of the paper to confirm and enhance the findings. Lastly, the results (presented in Section 5) were demonstrated and supported by quotations to enhance the study's reliability.

4.4 Validity Criteria

This section discusses the quality criteria adopted to ensure the rigor and trustworthiness of our qualitative investigation. Following the recommendations of Whitemore et al. [46] and Lenberg et al. [34], we evaluated our study against four key criteria: credibility, authenticity, criticality, and integrity.

Credibility. Credibility concerns the extent to which the findings accurately represent the meanings derived from the data, thereby instilling confidence in their truthfulness and interpretive soundness. To enhance credibility, multiple researchers were involved in the data collection and analysis phases, promoting analytical triangulation and reducing the influence of individual bias. In addition, to avoid potential bias arising from team association, external researchers participated in various stages of the study. All interpretations were substantiated with direct quotations from

player reviews, providing readers with transparent evidence that our insights were empirically grounded.

Authenticity. Authenticity, while closely related to credibility, emphasizes the researcher's ability to avoid influencing data collection and interpretation. In our case, data were obtained from publicly available video game reviews, ensuring that the dataset was not shaped or affected by researcher intervention. As such, both the collection process and the analytical procedures preserved the authenticity of participants' voices, thereby addressing this criterion alongside credibility.

Criticality. Criticality refers to the researcher's capacity for reflexive and systematic inquiry, characterized by the continuous questioning of assumptions, exploration of alternative interpretations, and recognition of potential biases. In this study, criticality was achieved through an iterative and reflective analytical process that combined a deductive theoretical framework with empirical validation via Leximancer. The refinement of the codebook, the manual inspection of concept relationships, and the reassessment of data patterns fostered a critical appraisal of emerging themes and mitigated the risk of uncritical verificationism. This recursive approach strengthened the interpretive rigor and ensured a nuanced understanding of the ethical dimensions present in player reviews.

Integrity. Integrity concerns the researcher's commitment to ensuring that interpretations remain transparent, data-grounded, and free from distortion or conjecture. In this work, integrity was pursued through methodological transparency, ethical data handling, and explicit documentation of each analytical phase. All interpretations were linked to supporting textual evidence, and the connection between theoretical constructs and empirical manifestations was systematically maintained. This rigorous and transparent process ensured that the reported findings were coherent, responsible, and faithful to the authentic perspectives expressed by players.

5 Findings of the Thematic Analysis—Responsibility for Ethical Issues in VGSE

Our thematic analysis process followed a deductive approach, where the coding was guided by a predefined framework of 19 ethical concern themes reported in subsection 3.2. These themes, derived from existing conceptualizations of ethical issues in video games, were used as analytical lenses to classify and interpret user quotations, i.e., video game reviews.

Across the corpus, we mapped a total of 170 quotations onto these themes, resulting in an average of 8.94 quotations per theme. Each quotation was further examined to extract the underlying key points, i.e., recurring aspects or sub-components of the ethical concern discussed in the review. In total, we identified 66 distinct key points, corresponding to about three to four key points per theme on average. By reviewing the key points, they were subsequently positioned along two complementary analytical concepts, namely *Responsibility* and *Lifecycle*, that naturally emerged during the analysis. The *Responsibility* dimension was directly elicited from the key points identified, as it represents the root of the ethical issue. It groups concerns according to the main actor and activity responsible for the identified ethical issue, distinguishing five categories: Developer Design Practice, Developer Accountability, Developer Governance, Player Misconduct, and Player Behavior. The *Lifecycle* dimension, instead, is a result of our interpretation of the results, and it represents *where* the issue should be addressed, guiding the implications of our study. It situates the same ethical concerns within six stages of the video game lifecycle, i.e., Design & Development, Pre-Release Governance & Policy, In-Game Interaction & Gameplay, Community & Social Interaction, Post-Release Moderation & Support, and Data, Privacy & Systemic Governance. Overall, this mapping highlights how the theoretical framework of ethical concerns finds empirical confirmation across multiple actors and stages of the video game ecosystem, with particularly frequent mentions

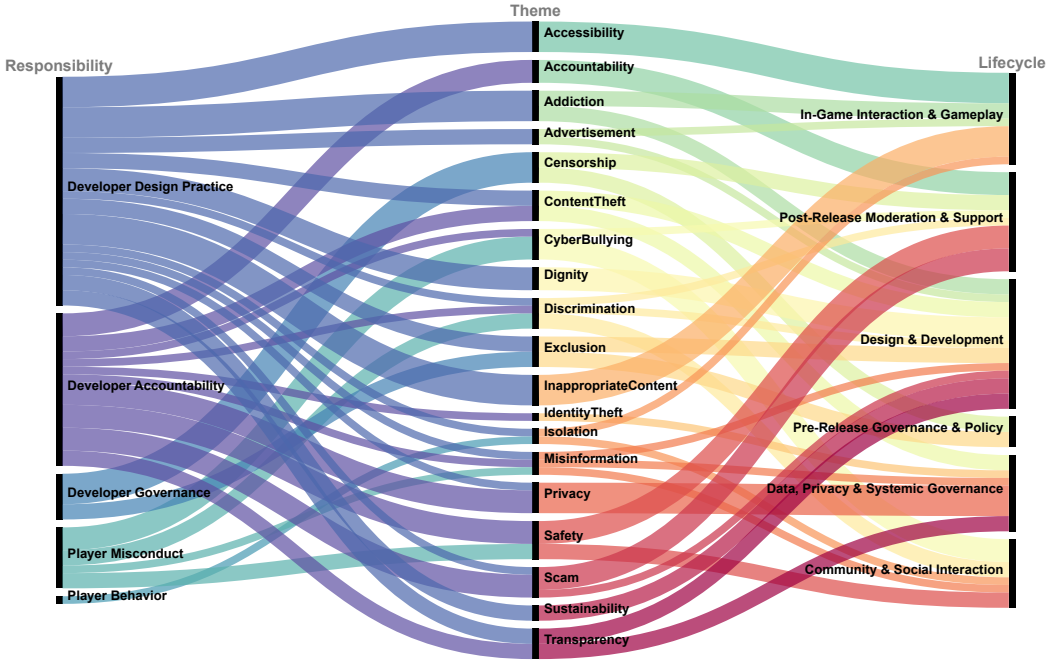


Fig. 3. Themes and relationships between them resulting from qualitative analysis.

for broader themes such as Censorship (13 quotations), Accessibility (11), and Cyberbullying (10), while other concerns such as Advertisement (2), Identity Theft (3), or Sustainability (6) appeared less frequently but remained more conceptually focused on specific systemic issues.

In the following sections, we discuss each theme organized along the *Responsibility* dimension, illustrating how these themes manifest in player reviews. We defer the discussion around the *Lifecycle* dimension to our Discussion section, as it is more relevant to our interpretation of the implications of our work than to the qualitative data itself.

5.1 Developer Design Practice

This category emerged as the most prominent source of ethical responsibility, capturing how developers' design choices shape fairness, inclusivity, and player well-being. Across the data, players linked ethical concerns to the mechanics, interfaces, and narratives of games, revealing design as a moral dimension where accessibility, representation, and monetization practices become ethically charged.

Accessibility. Players frequently pointed out that **many games fail to accommodate diverse needs and abilities**. Their quotations referred to issues such as rigid controls, limited customization options, and the absence of essential accessibility features like color-blind modes or adaptable interfaces. As one reviewer observed, *“all of the charm is unfortunately overshadowed by a lack of accessibility options and game-play that excludes those with hearing related disabilities”* (R1). Beyond specific functional barriers, these accounts reveal a broader sense of exclusion, where accessibility is perceived as an afterthought rather than a design priority. This perception emerges clearly in comments such as *“you cannot change the player movement key binds [...] but I am*

one-handed so I cannot change the movement keys to my foot switch" (R2), illustrating how the absence of customizable inputs directly prevents certain players from participating. Ultimately, these reflections position accessibility not merely as a usability concern but as an ethical imperative tied to **inclusion and equal participation in digital play**.

Addiction. Many reviews expressed frustration toward **design mechanisms perceived as intentionally addictive**. Players described reward systems, loot boxes, and microtransactions as psychological traps that exploit impulsive behaviors and foster dependency, as illustrated by comments such as ♡ *"the developers apparently had all the time in the world to build a working cash shop microtransaction system and financial transaction back end into the game. That shows the developers primary goal was just to set out to gouge cash from Steam gamers"* (R3). References to grinding, fan service, and endless loops further conveyed a sense of manipulation, where engagement metrics appear to outweigh player enjoyment. This perception emerges strongly in reviews like ♡ *"Everything is geared towards gambling addicts getting pulled in by pretty girls with squeaky voices and the monetization is obviously perdatory. Gacha games are gambling and gambling has no part in videogames"* (R4), where players explicitly link addictive design to exploitative monetization and affirm that **gambling has no part in video games**. Overall, the ethical critique here lies in the tension between commercial objectives and player autonomy: design decisions intended to sustain engagement risk transforming entertainment into compulsion.

Advertisement and Misinformation. Although less frequently mentioned, concerns related to advertisements and misinformation reflect players' growing awareness of the **manipulative and commercial dimensions of game design**. Many complained about the presence of promotional pop-ups, irrelevant cross-advertising, and monetization strategies that disrupt the sense of immersion, as illustrated by ♡ *"And the ads... oh my god, the ads. [...] You barely finish a match and suddenly you're bombarded with popups about unclaimed coins [...]"* (R5). Such remarks portray advertising not merely as an occasional annoyance but as a deliberate design choice that undermines authenticity and flow. Beyond commercial intrusiveness, some players also drew attention to forms of political or ideological propaganda embedded in games or their distribution, as shown in ♡ *"This violates multiple regional laws: – Article 436-2 and 109 of the Ukrainian Criminal Code (prohibition of glorifying aggression), – EU Digital Services Act (DSA) on harmful disinformation and military propaganda, – §130 of the German Criminal Code (incitement to hatred). Steam should not be a platform for state-sponsored militarization of youth"* (R5). Such statements reveal a broader concern for how digital platforms and games may inadvertently **propagate propaganda or harmful narratives, blurring the boundary between entertainment and information**. Through these accounts, players express unease at the commodification and instrumentalization of their attention—whether for profit or ideology—expecting developers and distributors to act responsibly toward both economic and informational integrity.

Content Theft. Some players raised ethical concerns about creative ownership and integrity in game design, particularly when developers were accused of **reusing community-created material without acknowledgment**. As one reviewer noted, ♡ *"I love TBOI and I dont care about the balance changes. But they blatantly stole a mods code, not the idea, not the art but the entire code. It was just stolen with no credit or changes"* (R6), highlighting a perceived breach of trust between creators and developers. Others criticized the monetization of user-generated content without fair attribution or compensation, as reflected in ♡ *"[...] Like another review said you are using user generated content so it makes sense to price it lower than 30 shmeckles"* (R7). Across these accounts, players framed creative reuse not as a technical or legal issue, but as a moral one—where **recognition and respect for contributors become essential to ethical game development**.

In their view, design practices that appropriate community work without credit compromise not only authenticity, but also the collaborative spirit that sustains modern gaming ecosystems.

Discrimination and Exclusion. Reviews within these themes emphasize inequality and representational imbalance in the gaming experience, referring to how design and market choices often reinforce unfairness, particularly through **economic and geographical barriers**. Some pointed to pay-to-win systems that privilege monetary investment over skill, as illustrated by Q “*is very f2p unfriendly. - Weapon system is entirely RNG. Your level, RNG, your quality, RNG. It is unfair, it is not acceptable*” (R8). Others criticized geographically restricted content, questioning why access to the same game experience depends on region, as reflected in Q “*The region lock is also awful. Why cant every country in the world experience the game?*” (R9). A further source of frustration concerned the lack of representation of certain regions and cultures, captured in statements such as Q “*Four out of seven continents are not or barely represented here. [...] Australia and the Asia-Pacific/Oceanic region, as well as Africa, are completely unrepresented*” (R10). Taken together, these accounts portray a **design ecosystem that reproduces existing social hierarchies—economic, cultural, and geographic—through design decisions**.

Inappropriate Content and Dignity. Players were often critical of excessive violence, sexualization, or disturbing scenes perceived as unnecessary to the narrative or gameplay, as illustrated by Q “*That’s kind of disturbing, for no good gameplay reason I found.*” (R11). Many quotations expressed disappointment with design choices that seem to prioritize shock or spectacle over coherence and audience respect, as noted in Q “*[...] there’s even a nudity setting. It doesn’t add anything interesting or meaningful, it’s just... there*” (R12).

Furthermore, players expressed how game content undermined people’s dignity through humiliation Q “*[...] These events are brutal, repetitive by design, and focused on humiliation and the physical consequences of abuse*” (R13), sometimes feeling an overall lack of humanity Q “*[...] Where exactly can we find humanity in this game?*” (R14). Together, these accounts highlight the tension between creative freedom and ethical responsibility: while developers may defend such elements as artistic expression, **players call for greater sensitivity to tone, context, and the potential for harm**.

Isolation. Several reviews linked feelings of loneliness and detachment to design limitations that restrict opportunities for meaningful social interaction. Players lamented the absence of cooperative or community features, interpreting this not merely as a gameplay limitation but as a missed opportunity to foster human connection, as captured in Q “*[...] funny game, please add a server list, since I have no friends and I am lonely*” (R15). These reflections highlight how the ethical concern stems from the relational dimension of play: **design choices not only shape mechanics and performance, but also the sense of belonging and connection that players derive from the gaming experience**.

Privacy, Sustainability, and Transparency. Several quotations addressed broader systemic aspects of design that extend beyond immediate gameplay. Players voiced concerns about unclear data policies, as in Q “*the supposed opt-out option thus acts as a placebo and misleads users about the actual state of data processing*” (16), and criticized misleading monetization strategies, exemplified by Q “*The new developer’s monetization policy immediately destroyed the attempts to find balance between free and paid players*” (17). Others mentioned technical inefficiency leading to overheating or excessive energy consumption, as described in Q “*On top of that, performance is rough: frame drops, stuttering, crashes, and overheating on both PC and consoles make the game feel unfinished.*” (18). Together, these reflections position developers as ethically responsible not only for user experience but also for environmental and informational impacts. **Design becomes an extended form of responsibility that spans usability, sustainability, and honest communication**.

5.2 Developer Accountability

This category captures how players interpret the absence of post-release responsibility as an ethical failure. They expressed frustration with unresponsive support systems, lack of moderation, and opaque communication, perceiving these gaps as forms of neglect rather than isolated technical issues. Whether the problem concerns unresolved bugs, unsafe environments, privacy violations, or discriminatory behavior, players consistently frame accountability as a moral expectation: developers are not only responsible for creating a functional product but also for maintaining fairness, safety, and transparency throughout its lifecycle.

Accountability. Across multiple games, players complained about the inability to report bugs, unethical behavior, or policy violations, as illustrated by Q “[...] *You guys are doing tournaments but not fixing your game? Good job! I just found a cheater but hey, I can’t report him*” (R19), as well as about the lack of responses when reports were submitted, as expressed in Q “*So many crashes the game is borderline unplayable. Reached out to customer support and no answer*” (R20). Together, the quotations convey deep frustration with opaque communication channels and the perception that companies abandon their users after release. Accountability here emerges as both a functional and moral expectation: **players want to be acknowledged and supported, and the absence of communication is interpreted as neglect and indifference.**

Content Theft and Misinformation. Players occasionally accused developers of reusing concepts or assets without proper acknowledgment, as seen in Q “[...] *The ‘developers’ here didn’t credit any of the artists and real, actual game developers who created the assets here, so this is plagiarism, as well as unethically made shovelware*” (R21), or of providing misleading information in promotional material and in-game descriptions, as expressed in Q “[...] *spreading misinformation about the viability of Remote Play in comparison to a native online mode helps nobody.*” (R22). Together, these accounts reveal a loss of trust in the creative and communicative integrity of the development process. They underscore the need for **transparency and accuracy in crediting, representation, and information sharing**—qualities that players perceive as essential to maintaining ethical standards in game production.

Cyberbullying, Safety, and Scam. Players frequently linked unsafe environments, toxic communities, and fraudulent behavior to insufficient developer intervention. Their accounts referred to scams in in-game economies, as in Q “*So in the end if you buy something in this game and you got unlucky like myself you’ll just end up being robbed with no way of getting your money back.*” (R23). Also, harassment and bullying within online chats were also recurring complaints, exemplified by Q “*Oh, but if you actually suffer active harassment inside the game, nothing will be done about it. Vibes-based moderation*” (R24), where the absence of moderation is interpreted as indifference toward victims. Similarly, several reviews reported that harmful or aggressive content is routinely tolerated, as captured in Q “[...] *Public chat frequently devolves into intense arguments and doxxing attempts, which moderators consistently ignore*” (R25). Although such **incidents often originate within the player community, reviewers ultimately hold developers accountable for enabling** them through inadequate tools, inconsistent rule enforcement, or the absence of transparent moderation policies. Across these accounts, safety emerges as a fundamental ethical expectation.

Privacy, Transparency, and Identity Theft. Concerns related to data protection and transparency emerged prominently when players discussed intrusive permissions, unclear governance, and the risk of compromised accounts. Several reviews accused developers of violating personal boundaries, as in Q “*The developer and their administrators pry into players’ private data without consent*” (R26). Players also questioned the honesty of marketing and disclosure practices, as captured in Q “*The*

game isn't marked as early access on the shop page [...] Misleading? Maybe. Illegal? Probably. But hey, who needs transparency when you've got potential, right?" (R27) Finally, the risk of account theft was a recurring anxiety, illustrated by *Q* *"I also dislike their account system, since giving away the ID of your account is half way to getting hacked"* (R28). Across these quotations, players frame privacy as part of a moral contract between user and developer—one that demands respect, transparency, and care in handling personal data. Ultimately, **trust becomes the central ethical issue**: it is undermined not necessarily by intentional wrongdoing, but by systemic opacity, weak safeguards, and poor communication and support.

Discrimination. Lastly, several reviews described situations in which developers themselves were perceived as biased or unjust in their treatment of players. Some accounts accused development and moderation teams of favoring certain groups or discriminating against others, as illustrated by *Q* *"Misogynistic and transphobic dev and moderation team, who don't give any fucks about discrimination but if you try to stop it or defend someone they ban you"* (R29). Such experiences suggest that discrimination is not confined to gameplay dynamics or community behavior, but may also emerge from within the structures meant to ensure fairness and inclusion. These perceptions generate a strong sense of betrayal and unfairness, reinforcing the notion that **accountability extends beyond technical maintenance to the equitable governance of player relations and community discourse**.

5.3 Developer Governance

This category captures ethical concerns rooted in developers' and publishers' policy-level decisions that regulate what content is available and to whom. Players described governance practices such as content censorship and regional restrictions as mechanisms of control that shape fairness and inclusivity even before gameplay begins. Across the reviews, these decisions were perceived not only as commercial or legal strategies, but as moral choices that define the boundaries of creative expression and global participation.

Censorship. The theme of censorship emerged as a complex intersection of governance and accountability in game development. Players discussed both pre-release and post-release forms of control, highlighting how content governance and moderation policies shape their perception of fairness and creative integrity. At the pre-release level, many expressed disapproval of cultural censorship, as seen in *Q* *"Changing my review to negative. This patch added censorship, changed story, and removed entire combat encounters. These changes were made to comfort Chinese nationalism."* (R30), and of content cuts implemented to satisfy market expectations or regional sensitivities, as illustrated by *Q* *"They are censoring content we paid for to appease a bunch of lunatics for their cOnSoLe release. The review will remain negative until everything censored is reverted. Stay away from shady practices like these. Censorship is poison."* (R31). These players perceived such interventions as compromising artistic expression and disrespecting their audience, revealing a **tension between commercial prudence and creative authenticity**.

Post-release, the same concern took the form of frustration toward moderation practices and opaque communication. Players reported unjust bans and unexplained restrictions, interpreting these as evidence of inconsistent or biased enforcement, as described in *Q* *"[...]the devs often get into player affairs personally, having permanently banned several users for some pretty mundane stuff. [...] when confronted by the EU following a GDPR investigation, they said they had no evidence against them"* (R32). In this phase, censorship overlaps with accountability: **the issue is not the existence of rules, but the lack of transparency and fairness in how they are applied**. While players may accept certain forms of governance to ensure compliance or community safety, they

consistently demand ethical clarity—expecting developers to exercise control with transparency, consistency, and respect for audience autonomy.

Exclusion. Players also discussed exclusion as a governance-level issue, often pointing to region locks and other restrictive management policies that limit access to certain markets or networks. Some described these practices as unclear or poorly communicated, as in ♫ *“Some people are mentioning game is region locked but there’s no official statement about this either. If you going to region lock a game at least mention it somewhere”* (R33), while others emphasized their direct impact on marginalized regions, as illustrated by ♫ *“As someone from Iran, I face constant regional restrictions [...] And I’m not alone—many players in restricted regions are locked out due to the game’s heavy-handed DRM and unreliable peer-to-peer matchmaking”* (R34). Across these accounts, players **framed region-based restrictions not simply as technical constraints but as ethical barriers that divide global communities and reinforce geographic inequality**. Such governance decisions therefore shape fairness before players even enter the game, determining who is allowed to participate and under what conditions.

5.4 Player Misconduct

This category captures ethical concerns arising from harmful behaviors within player communities. Players described toxic exchanges, discrimination, harassment, and deceptive practices that compromise fairness and safety in multiplayer environments. Unlike issues rooted in design or governance, these concerns originate from peer interactions, revealing how the ethical landscape of gaming extends beyond developers’ actions to include the culture and conduct of players themselves. Such accounts underline that fairness and respect are collective responsibilities—sustained not only by technical moderation, but by the everyday choices and attitudes of the community.

Cyberbullying. Many quotations described toxic exchanges and harassment within the community, as illustrated by ♫ *“Voice chat is frequently misused, with players encountering verbal abuse and xenophobic remarks. The tools exist, but the sheer volume of incidents suggests a need for stronger moderation and better matchmaking filters”* (R35). Players frequently mentioned how these behaviors disproportionately affected newcomers, who often became targets of aggression and exclusion, as captured in ♫ *“Unfortunately the game is plagued with Unfairness. New players cannot play as the veteran players go bully the newcomers in the first map with high tier gear, (there is no restrictions) So you can’t even start enjoying the game”* (R36). Such dynamics not only harm individuals but also erode the sense of belonging within gaming communities, creating environments where participation becomes intimidating or discouraging. Players explicitly recognized the responsibility of developers in implementing effective moderation systems, yet framed cyberbullying primarily as a community-driven ethical failure—one that requires not only technical intervention but also a **shift in player culture toward empathy and respect**.

Discrimination and Safety. Players frequently reported experiences of social discrimination and threats to personal safety within gaming communities. Instances of racism, sexism, and other forms of prejudice were described as pervasive and normalized, as seen in ♫ *“it takes in consideration people as logical being, while they are not. Most of the cases are pure racism, sexism [...]”* (R37). Alongside these concerns, many players expressed fear for their safety, emphasizing how unmoderated voice or chat systems expose them to harassment and abuse, as illustrated by ♫ *“The publisher doesn’t record voice chat, so you can be insulted in every way and nothing happens to those players. This creates a toxic community and an unsafe place for younger players”* (R38). Such accounts also highlighted the **vulnerability of younger audiences**, who may be particularly affected by toxic exchanges

and predatory behavior. Together, these experiences underline that the social dimension of play is deeply ethical: **freedom of interaction coexists with responsibility toward others, and its neglect transforms open communities into unsafe or exclusionary spaces.**

Misinformation. Ethical issues within player communities also included the deliberate or unintentional spread of false information, particularly through misleading or fake reviews of video games—a concern that players frequently identified as harmful to collective trust. As one reviewer noted, “*This game is honestly pretty great, but there is a ton of misinformation in the reviews here on Steam*” (R39), emphasizing how distorted narratives can shape public perception and unfairly influence reputations. Such practices were denounced as violations of mutual trust, revealing that the **ethical economy of games extends beyond developer–user relations** to the credibility and responsibility of peer-to-peer interactions.

5.5 Player Behavior

This category highlights ethical reflections that stem from individual player experience rather than interpersonal misconduct. While less frequent, these accounts reveal how personal emotions, habits, and attachments can themselves become ethical concerns.

Isolation. Finally, some reviews reflect on individual experience rather than collective harm, focusing on how games can both accompany and intensify feelings of loneliness. Players described how prolonged engagement sometimes led to emotional detachment or solitude, as expressed in “*This is a great game, but make sure you have friends or else it get lonely up here...*” (40), or even to a form of dependency on virtual companionship, as in “*When I first went out with Seung-Yi I felt more lonely than the times I had panic attacks in university because I failed to build good relationships with my classmates and I was close to being expelled due to the piling difficulties that come with being alone. It’s mortifying how deeply comforting it was to go out with her*” (41). Unlike other categories, these reflections convey self-awareness and moral introspection rather than external accusation, suggesting that **ethical concerns in gaming can also emerge through players’ recognition of their own emotional vulnerabilities** and the blurred boundary between connection and isolation.

6 Discussion and Implications

In this section, we present and discuss our results through the lens of the second analytical perspective introduced in our work, namely, identifying *when* the ethical concerns emerging from player reviews should be addressed along the game lifecycle, as depicted in Figure 3. By situating the themes identified in our analysis within their temporal and procedural context, we highlight at which stage ethical responsibility becomes most salient and how it should be operationalized in practice.

As an additional contribution, we present our experience using Leximancer to analyze software engineering artifacts. Leximancer remains relatively unknown in our community, and we argue—based on our experience—that this represents a missed opportunity that should be addressed. We hope that our report will encourage its adoption.

6.1 Implications for Video Game Developers and Users

Building on this temporal framing, we derive implications for three key actors in the videogame ecosystem: (1) *game maintainers*, who are responsible for ethical governance and player well-being in ongoing productions; (2) *game developers*, who can embed fairness and inclusivity into

the design and development of new titles; and (3) **players**, whose behavior and community norms play a crucial role in sustaining ethical interaction within digital spaces.

Design & Development. Ethical concerns belonging to the design and development stage reveal how core creative and technical choices embed moral assumptions into the game's architecture. Players discussed **Addiction** mechanisms, such as manipulative reward loops, microtransactions, and gacha systems, that exploit impulsive behavior rather than foster enjoyment. They also criticized intrusive **Advertisement** and ideological **Misinformation (Propaganda)** as deliberate design strategies that commodify attention and shape perception beyond the game itself. Concerns about **Content Theft** underscored the importance of crediting modders and community contributors, while reports of **Discrimination**, **Dignity**, and representational **Exclusion** pointed to how characters, stories, and aesthetics can reproduce social hierarchies or cultural bias. Finally, issues such as **Sustainability**, misleading **Transparency**, and unclear **Privacy** practices highlighted the ethical dimension of performance, data collection, and communication during design.

🔧 For *game maintainers*, this means revisiting existing mechanics, particularly monetization and engagement systems, to ensure they support autonomy and well-being rather than compulsion. Maintainers should also communicate transparently and clearly about in-game economy adjustments and the reuse of community content, avoiding silent policy changes that erode trust.

</> For *game developers*, ethics should be integrated as a concrete design constraint: applying fairness and inclusivity audits to representation, conducting code and asset provenance checks to prevent appropriation, and planning for resource efficiency and transparent data handling from the earliest stages.

Ultimately, fairness in design is achieved not through corrective patches but through proactive alignment between creativity and responsibility. When ethical awareness is treated as a design parameter, alongside usability or performance, games can better balance innovation, inclusivity, and long-term sustainability.

Our observations on manipulative engagement loops and monetization mirror ethical concerns previously identified in the context of gambling-like game mechanics [9, 23, 31]. Likewise, issues of representation and dignity echo prior evidence that video games often reinforce gender and racial stereotypes [6, 35, 39], extending these findings to user perceptions of fairness in design.

Tjikhoei et al. [44] noted that addictive design and discrimination were among the most frequently reported ethical concerns across mobile applications. Our findings extend this evidence by showing how, in the gaming context, these same concerns are intertwined with creative intent and player immersion, rather than limited to app-level usability or monetization. Moreover, Li and Aslam [36] observed that prejudice and exclusion are widely discussed on Reddit, mirroring our findings on representational imbalance and reinforcing that player communities recognize bias in both content and system design.

Community & Social Interaction. Once a game becomes a social space, ethical concerns extend beyond design and enter the domain of player conduct and community norms. Players described widespread **Cyberbullying** and harassment, highlighting how toxicity, exclusion of newcomers, and verbal abuse in chats undermine trust and belonging. Accounts of **Discrimination** and violations of **Dignity**—through racist, sexist, or degrading behavior—reveal how offline social hierarchies are reproduced within virtual communities. Concerns about **Safety** and scams, including stalking, doxing, and fraud in player-to-player trades, further illustrate that unmoderated environments can become unsafe or exploitative. At the same time, peer-level **Misinformation** spread through fake reviews or manipulated narratives damages collective credibility, while some players reflected on

Isolation within toxic communities, perceiving social interaction as emotionally draining rather than enriching. Finally, **Content Theft** between players—such as uncredited reuse of mods or in-game creations—was interpreted as a breach of mutual respect and authorship ethics.

Mitigating these issues requires fostering community environments that promote empathy, accountability, and transparency.

- 🔧 For *game maintainers*, this means going beyond technical moderation and investing in ethical community design: implementing clear codes of conduct, responsive reporting systems, and public moderation logs that communicate enforcement rationale. Encouraging community-driven moderation and recognizing positive behavior can also strengthen social resilience against harassment and deceit.
- 🔗 For *players*, these findings emphasize their dual role as participants and stewards of digital ethics. Toxicity, discrimination, and misinformation are not only failures of platform control but reflections of cultural norms that players can challenge through example, peer accountability, and collective action.

Ethical interaction, therefore, emerges not from stricter rules alone but from a shared moral infrastructure, where fairness, respect, and safety become the community's own values rather than externally imposed constraints.

Extending Li and Aslam [36]'s insight that Reddit communities actively negotiate moral boundaries through public discourse, our findings emphasize how toxicity, misinformation, and discrimination transform the game space into a moral arena. These dynamics corroborate Tjikhoeiri et al. [44]'s identification of cyberbullying as a user-defined ethical category, demonstrating that community self-governance is an essential dimension of fairness in interactive environments.

In-Game Interaction & Gameplay. This stage concerns the ethical quality of players' direct engagement with a game, the moment where design decisions materialize into lived experience. Players frequently mentioned **Accessibility** as a fundamental yet neglected condition for fairness, pointing to missing control customization, absent subtitles, or a lack of features for players with disabilities. When addressed, accessibility is often treated as an optional enhancement rather than an ethical obligation. Reports of in-play **Addiction** revealed how engagement loops and fan-service dynamics can transform enjoyment into compulsion, particularly when rewards and attention cues are engineered to sustain constant play. Intrusive **Advertisement** during gameplay amplifies frustration, as pop-ups and monetized prompts disrupted immersion and blurred the line between play and promotion. Players also criticized **Inappropriate Content**—including gratuitous violence or sexualization—for undermining narrative coherence and unnecessarily alienating parts of the audience. Meanwhile, some reflected on **Isolation**, describing how repetitive solo play or the lack of cooperative features fostered emotional fatigue rather than connection.

At this level, most concerns can be mitigated through user-centered testing.

- 🔧 For *game maintainers*, this involves monitoring live player feedback to identify accessibility gaps, unhealthy engagement patterns, or content perceived as disrespectful or alienating. Adjusting in-game events, balancing reward systems, and adding inclusive interface options are concrete ways to maintain ethical consistency after release.
- </> For *game developers*, these findings emphasize the need to treat accessibility, moderation of addictive feedback, and cultural sensitivity as baseline design requirements rather than reactive fixes. Prototyping with diverse testers and integrating accessibility and well-being checkpoints into playtesting cycles can ensure that fairness and inclusion remain embedded within the game's experiential core.

Ultimately, ethical gameplay arises when interaction design respects users' cognitive and emotional limits, enabling agency, inclusion, and meaningful connection instead of dependency or exclusion.

Concerns about gratuitous violence and sexualization echo long-standing debates on the moral responsibility of developers in shaping narrative tone and player affect [2, 12]. Beyond this, players' introspective reflections on loneliness and emotional fatigue resonate with Sicart's view of games as moral laboratories, where self-awareness emerges through play [25, 41].

Post-Release Moderation & Support. After release, ethical responsibility shifts toward maintaining fairness, safety, and transparency in the ongoing relationship between developers and their communities. Players repeatedly voiced frustration about absent or ineffective **Accountability**, describing situations where bug reports, cheating complaints, or user issues received no acknowledgment or resolution. Concerns over **Safety** were equally prominent, with players noting the lack of consistent moderation, vague enforcement policies, and slow responses to harassment or harmful content. Unclear or arbitrary **Censorship**, including unexplained bans or deleted posts, was seen as further evidence of weak communication and discretionary power. Finally, some reviews exposed perceived **Discrimination** in support and enforcement practices, where certain groups or opinions were reportedly treated unfairly. These accounts collectively underscore that ethical failures at this stage are not rooted in technical incapacity but in the erosion of procedural trust.

✂ For *game maintainers*, it means they must treat support and moderation as extensions of design ethics rather than peripheral services. Transparent ticket-tracking, public moderation reports, and clear escalation channels can reduce perceptions of arbitrariness. Investing in trained moderation teams, equipped to detect bias, communicate clearly, and act consistently, helps sustain fairness across player interactions. Automated enforcement should be complemented by human oversight to ensure that context-sensitive cases, such as hate-speech reports or wrongful bans, are handled equitably.

Ultimately, post-release ethics depend on the continuity of care. When moderation and support are treated as living components of the game ecosystem—open, responsive, and equitable—trust becomes a sustained outcome rather than a temporary promise.

Similarly to how reported by Tjikhoei et al. [44], accountability remains a central expectation from users, who demand consistent moderation and transparent support systems. Yet, in games, the persistence of post-release issues indicates that ethical responsibility does not end at deployment—it requires continuous governance mechanisms embedded in live operations.

Pre-Release Governance & Policy. Ethical concerns emerging at the pre-release stage highlight how governance and market decisions can determine fairness before players even access a game. Themes such as **Censorship** and **Exclusion** show that choices about content visibility and regional availability are not merely logistical but moral in nature. Players criticized content removal and cultural censorship as violations of creative integrity, seeing them as commercially driven compromises that erode artistic authenticity and disrespect local audiences. Similarly, region locks and geographically differentiated pricing were perceived as exclusionary practices that reproduce global inequalities and restrict participation based on nationality or purchasing power. Addressing these issues requires governance processes that are both transparent and participatory.

✂ For *game maintainers*, this means openly communicating the rationale behind policy decisions, providing pre-release notes that explain content adjustments, and maintaining changelogs accessible to players across regions. They should also establish consistent ethical review pipelines to ensure that modifications driven by regional or political constraints are documented, justified, and reversible when possible.

</> For *game developers*, early design and publishing negotiations should explicitly consider cultural and accessibility impact, adopt cross-regional testing, consult with local communities, and plan flexible release strategies that avoid the need for post-hoc censorship or exclusion.

Our findings on cultural censorship and exclusion resonate with the community-level concerns identified by Li and Aslam [36], who found similar frustration among players about region-specific restrictions and politically motivated content adjustments. These early governance choices, as also reflected in Wright's notion of justice Wright [49], illustrate how fairness must be considered from the earliest stages of production.

Data, Privacy & Systemic Governance. At the systemic level, ethical concerns extend beyond gameplay and moderation to the invisible infrastructures that govern data, ownership, and communication. Players expressed anxiety about unclear **Privacy** policies and consent mechanisms, criticizing developers who collect or share personal information without transparency or meaningful opt-out options. Similarly, the lack of **Transparency** in how data are processed, monetized, or secured eroded user trust, with some reviews describing such opacity as deceptive rather than merely negligent. Accounts of **Identity Theft** and weak account protection revealed fears of exploitation that transcend individual games, affecting players' broader digital identity. Finally, concerns about **Content Theft** and the systemic spread of **Misinformation** pointed to the ethical responsibility of platforms to protect creative ownership and maintain information integrity across interconnected ecosystems. Addressing these issues requires an ethical infrastructure that operates at both organizational and technical levels.

</> For *game developers*, the implication is that ethical data management must be designed into the system itself: privacy-by-design architectures, explainable data flows, and verifiable attribution mechanisms can prevent misuse before it occurs. Such technical safeguards should be complemented by organizational ethics—data audits, bias evaluations, and public commitments to information accuracy—to sustain long-term player trust.

As noted by Li and Aslam [36], ethical debates around games increasingly involve systemic issues—privacy, data exploitation, and trust in platform governance—rather than only content or conduct. These align closely with the foundational ethical dimensions of autonomy and transparency in Wright's EIA framework [49], showing continuity between broader IT ethics and domain-specific gaming ecosystems.

6.2 Future Research Directions

This study opens several promising directions for future research within the VGSE community. In this section, we reports some of them and discuss in details.

Our findings demonstrate that ethical aspects can indeed be detected and analyzed through player reviews, highlighting their potential as a valuable and underexplored data source for understanding moral and ethical dimensions in video game development. ♡ **Consequently, researchers should explore the design and implementation of automated tools capable of identifying ethical issues directly from player-generated content.** Such tools could combine advances in natural language processing, sentiment analysis, and ethical reasoning models to support developers and researchers in continuously monitoring player perceptions and detecting early signals of ethical risks. We consider this a central implication of our work, as it could foster a new generation of data-driven systems for ethics-aware SE in the gaming domain and also because a similar avenue has already been explored for app reviews [13, 14, 26].

Nevertheless, we argue that this entire research agenda suffers from a concrete risk: everyone lies, also in app reviews. Not all reviews reflect truthful or unbiased perspectives—indeed, manipulation

and misinformation are widespread phenomena in online environments [37]. The practice of *review bombing* is a prominent example of how collective behaviors can distort public perception and inject misleading signals into user-generated datasets [5]. These distortions raise two intertwined research challenges. ¶ **First, there is a clear need to develop robust methods for detecting and filtering deceptive or biased reviews** through hybrid approaches that combine linguistic, behavioral, and network-based indicators. ¶ **Second, further work should investigate how misinformation and deception influence software development processes and developers' decisions**, to understand their impact on design choices, reputation management, and ethical responsibility; we argue that this could be an impactful contribution to the human and social aspects of the research community (VG)SE, especially seeing the great impact review bombing has on the market [5].

Moreover, our proposed framework should be regarded as both complementary to and an extension of the work of Li and Aslam [36] discussed in Section 2. While that work offered a qualitative exploration of ethical issues in games, our contribution advances this line by extending the framework and empirically validating and structuring these concepts through large-scale data analysis. ¶ **Future research could further enhance this perspective by integrating developer-oriented studies—both qualitative and quantitative—that explore how practitioners perceive, prioritize, and address ethical concerns emerging from player feedback.** Combining the viewpoints of players and developers would yield a richer and more comprehensive understanding of ethics in VGSE, ultimately supporting the creation of actionable guidelines and design principles for responsible game development.

Finally, like all empirical investigations, our study has several limitations, detailed in Section 4. Addressing these limitations represents another avenue for future research. ¶ **We encourage follow-up studies to replicate and extend our approach across different platforms, genres, and cultural contexts, to test the generalizability of our findings and to refine the framework accordingly.** By doing so, future research can either strengthen or challenge the theoretical and empirical foundations laid by this work, contributing to a more mature and systematic understanding of ethical issues in video game software engineering.

6.3 On the Use of Leximancer

Our initial objective was to conduct a large-scale, semi-automated content analysis of video game reviews by relying on Leximancer's automated concept extraction and quantitative reporting features. The plan was to use the tool to identify concepts and co-occurrence statistics across hundreds of thousands of reviews, thereby obtaining a quantitative overview of the most salient ethical themes emerging from player discourse. However, as the analysis unfolded, it became evident that the results provided by Leximancer were not sufficiently reliable for drawing meaningful quantitative conclusions. This prompted a shift toward a more qualitative and interpretive use of the tool, as also suggested in the tool's guidelines [42].

Two factors likely contributed to this limitation. First, **the scale and heterogeneity of the dataset**—comprising more than 300 000 user reviews—introduced a high degree of lexical sparsity and noise. Because Leximancer's underlying thesaurus learning mechanism depends on word co-occurrence frequencies, the vast and unbalanced corpus may have hindered the model's ability to discern conceptually coherent terms, leading to overly diffuse concept maps and partial misclassifications. Second, **the linguistic nature of video game reviews** further complicated automated classification. Player discourse often involves irony, sarcasm, exaggeration, and the frequent use of humor or informal registers. In addition, ethically sensitive content—such as violence, addiction, or toxicity—is not necessarily criticized by players: as demonstrated by our qualitative analysis, it is often mentioned approvingly as a desirable or expected gameplay element, in some cases.

This ambiguity in evaluative stance makes it difficult for unsupervised semantic analysis tools to distinguish between positive and negative mentions of ethically charged content.

Recognizing these limitations, and in line with Leximancer’s intended use [42], we repositioned the tool from a primary analytic engine to a supportive methodological instrument within a broader qualitative workflow. Rather than relying on its quantitative outputs, we used Leximancer to facilitate the development and refinement of our codebook and to aid the targeted retrieval of illustrative quotations, as also done in previous literature [16, 20, 48]. Specifically, we exploited its concept mapping capability to explore thematic relationships and its indexing function to search the dataset efficiently for reviews exemplifying each ethical concern. This iterative integration allowed us to maintain a deductive perspective—anchored in our predefined ethical framework, while leveraging Leximancer’s computational strength to manage and navigate the vast corpus.

In retrospect, the experience provided two key methodological insights. First, when applied to highly contextualized and affective domains such as video game reviews, **automated content analysis** should be treated as a scaffolding mechanism for human interpretation rather than as a substitute for qualitative reasoning. Second, the value of large-scale data lies not in its complete manual analysis, but in how it informs the refinement of conceptual models through iterative human-machine interaction.

7 Conclusions

This work addresses the growing interest in ethical issues in video games from a design and product maintenance perspective. Our objective was to investigate whether player reviews could be used as a data source for identifying ethical issues and whether these could be systematically categorized to provide a more manageable representation of the problem. Our results not only demonstrate that this is possible, but also show that ethical issues can be—starting from the reviews—mapped to their sources (e.g., development teams, players, or design decisions) and linked to specific phases of the development lifecycle, thus supporting an engineering-oriented approach to managing such ethical concerns.

Our work contributes to the VGSE community in the following ways:

- We provide an ethical framework grounded in literature and developed through a collaborative research process, extending existing frameworks [36, 44, 49] into the entirely new context of video games.
- We conduct a large-scale evaluation of this framework on over 300 000 player reviews, offering a detailed analysis of ethical issues in terms of their sources and the development phases in which they emerge.
- We make all our data publicly available in our online appendix [45] to ensure open-science and reproducibility.
- We share our experience with Leximancer, a tool still rarely adopted in the Software Engineering community but which, in our view, offers interesting and valuable analytical capabilities.

As future work, we plan to conduct a quantitative study to examine developers’ perceptions of these ethical issues, followed by a qualitative investigation to explore key aspects in greater depth—thus adopting a mixed-method approach [7]. In parallel, we intend to begin developing agent-based tools capable of automatically collecting data and integrating into development workflows to help identify and address ethical concerns. We believe these steps will contribute to addressing a critical aspect of game development—one that also influences the public image and sustainability of development companies—thereby supporting both ethical awareness and productivity in the video game industry.

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- Conflict of interest: the authors declare that they have no conflict of interest.
- Ethics approval and consent to participate: no ethical approval was required since no human participant was involved. All data was collected from publicly accessible web pages in compliance with Steam's Terms of Use. No personal data was collected or inferred, and the scraping was performed responsibly, at low frequency and with user-agent identification, solely for academic research purposes.
- Data availability: the data that support the findings of this study are openly available in our online appendix [45].
- Author contribution: all authors contributed to the study conception and design. Material preparation, data collection and analysis were performed by Stefano Lambiasi and Gianmario Voria. The first draft of the manuscript was written by Gianmario Voria, and all authors commented on previous versions of the manuscript. All authors read and approved the final manuscript.

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